<TEST>

Nov. 23th, 2020

150 minutes / Closed Book

Calculator & 2-Page Summary Sheet Allowed

Problem 1 [10pt, 5pt each]

The following matrix displays the joint probabilities of different weather conditions and of different recreation benefit levels obtained from use of a reservoir in a national park.

Weather	POSSIBLE RECREATION BENEFIT LEVELS		
	RB1	RB2	RB3
Wet	0.10	0.20	0.10
Dry	0.10	0.30	0.20

(1) Compute the probabilities of recreation levels RB1, RB2, RB3 and of dry and wet weather.

(2) Compute the conditional probabilities P(wet|RB1), P(RB3|dry), and P(RB2|wet).

Problem 2 [15pt, 5pt each]

A random variable R has a triangular probability density function(pdf) $f_R(r) = 1\text{-}0.5r$ for $0< r <2$ and zero otherwise. However, the interest in not really in R, but the weight of particles given by $W=7R^3$.

(1) What is the mean and variance of R? Show your work.

(2) What is the pdf for W?

(3) Are your results could be difference if $0\leq r \leq 2$? Why?

Problem 3 [15pt, 5pt each]

Barges arrive at a lock at an average of 4 each hour.

(1) If the arrival of barges at the lock can be considered to follow a Poisson process, what is the probability that 6 barges will arrive in 2 hours?

(2) If the lock master has just locked through all of the barges at the lock, what is the probability he can take a 15-minute break without another barge arriving?

(3) If the operation of the lock is such that 4 barges can be locked through at once and the lock master insists that this always be the case, what is the probability that the first barge to arrive after 4 previous barges have been locked through will have to wait at least 1 hour before being locked though?

Problem 4 [10pt, 5pt each]

Verify the following sentences:

(1) The pdf of an extreme value random variable depends on the sample size as well as the parent distribution.

(2) $\lambda_2 = \sigma/(\pi)^{0.5}$ for the normal distribution where λ_2 is the second L-moment called 'the L-scale' and σ is the standard deviation of a random variable.

Problem 5 [15pt, 5pt each]

(1) Derive the Yule-Walker equation for a time series analysis.

(2) Derive the variance of the L-month ahead forecast of AR(1).

(3) Write a equation for a 4-month ahead forecast of ARMA(2,1).

~ *Good Luck!* ~